Deep-Learning-Based Cloud Masking on Multispectral Ocean Color Imagery for Floating Macroalgae Monitoring

Mingqing Liu, Mengqiu Wang, Yue Sun, and Zhongbin B. Li

***Abstract*— Intensive blooms of floating macroalgae have been widely reported during the past decades. Multispectral satellite imagery serves as an important data source for bloom monitoring, but its capacities are often hindered by the difficulties in accurate cloud or cloud shadow masking. Imperfect masking not only leads to a lack of bloom observations but also induces uncertainties in their biomass estimations. To reduce false detections and to retain more valid satellite measurements, the feedback attention network (FANet) was applied to mask cloud and cloud shadow pixels on Moderate Resolution Imaging Spectroradiometer (MODIS) imagery. Satisfactory performance was achieved on the images taken in the Caribbean Sea, where cloud and sun glint contaminations frequently occur. The overall cloud detection accuracy is ∼96%, as referred to manually prepared "ground truth". Cloud shadows are effectively detected while overmasking near sun glint regions is reduced. Compared with the commonly used SeaWiFS Data Analysis System (SeaDAS) cloud products, the FANet-derived cloud products retain 44% more valid MODIS measurements in summer 2021. Consequently, they consistently provide more macroalgal features on each single MODIS image. However, the monthly mean macroalgae biomass statistics derived from these two cloud masking approaches are consistently close despite their large differences in the number of valid measurements, suggesting that the mean-value composite strategy can minimize the impact of missing observations and provide reliable long-term trends. The proposed cloud masking method is also applicable to other similar multispectral sensors [i.e., Visible Infrared Imaging Radiometer Suite (VIIRS)] for monitoring floating macroalgae in the global open oceans.**

***Index Terms*— Alternative Floating Algae Index (AFAI), cloud detection, deep learning, ocean color, *Sargassum*.**

I. Introduction

FREQUENT macroalgal blooms have been observed in global oceans and caused severe negative consequences during subsequent beaching events [1], [2], [3], [4], [5]. Satellite remote sensing provides timely and effective information on the surface oceanic features, and thus has been used as a critical data source to monitor and track the blooms [6], [7], [8], [9]. Currently, multispectral ocean color sensors including the Moderate Resolution Imaging Spectroradiometer (MODIS), Visible Infrared Imaging Radiometer Suite (VIIRS), and Ocean Land Colour Instrument (OLCI) have been used for the detection of *Sargassum* blooms and provided valuable information on their large-scale distributions in the Atlantic Ocean [10], [11], [12], [13].

Optical sensors cannot see through clouds. In the Caribbean Sea and tropical Atlantic, frequent cloud cover leads to considerable amounts of invalid measurements on all the aforementioned sensors. Currently, SeaWiFS Data Analysis System (SeaDAS) quality flags have been widely used to mask the cloud or cloud shadow contaminated pixels. For examples, Ody et al. [14] used the SeaDAS CLDICE (clouds or ice) flag [15] to mask cloud pixels on MODIS and VIIRS when detecting the floating *Sargassum* aggregations in the tropical North Atlantic. In a recent study, the standard CLDICE flag was also used by Hu et al. [16] to identify cloud pixels on MODIS, VIIRS, and OLCI imagery to reduce the impact of cloud on pelagic *Sargassum* quantifications in the Atlantic Ocean.

However, the SeaDAS cloud mask is primarily designed to process high-quality global-wise ocean color data. When monitoring marine floating macroalgal blooms, it can misidentify many useful observations as noise, thus reducing valid observations. In fact, the default of CLDICE flag simply uses a threshold of reflectance at 869 nm greater than 0.027 to determine the cloud pixels [15]. As the spectral properties may vary in different regions and seasons, simple fixed thresholds are not necessarily applicable to cloud detection under various complex conditions. It is already noted that CLDICE may overmask valid cloud-free pixels and undermask cloud or cloud adjacent pixels [11]. Xing et al. [17] have demonstrated that CLDICE based on near-infrared (NIR) band [18] would mask floating macroalgae and lead to underestimation of the bloom abundance. An improved cloud masking approach N09CM [19] was designed based on CLDICE to reduce overmasking. The Floating Algae Index (FAI) and Alternative FAI (AFAI) products designed to detect floating algae can be relatively robust to diffuse clouds due to the band subtraction design [20]. Even with the presence of thin clouds, the red-edge feature can still be strong enough to detect the presence of floating macroalgae [20], making these pixels useful for bloom monitoring. Fig. 1 shows an example of the limitations of the SeaDAS CLDICE-based masking (SeaDAS_cloud) for floating algae detection on MODIS AFAI images, where overmasking near sun glint regions and failed detection of residual clouds or

Manuscript received 21 July 2023; revised 31 October 2023 and 21 November 2023; accepted 23 November 2023. Date of publication 30 November 2023; date of current version 12 December 2023. *(Corresponding author: Mengqiu Wang.)*

The authors are with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China (e-mail: mengqiuwang90@gmail.com).

This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2023.3337238, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2023.3337238

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:33:09 UTC from IEEE Xplore. Restrictions apply.

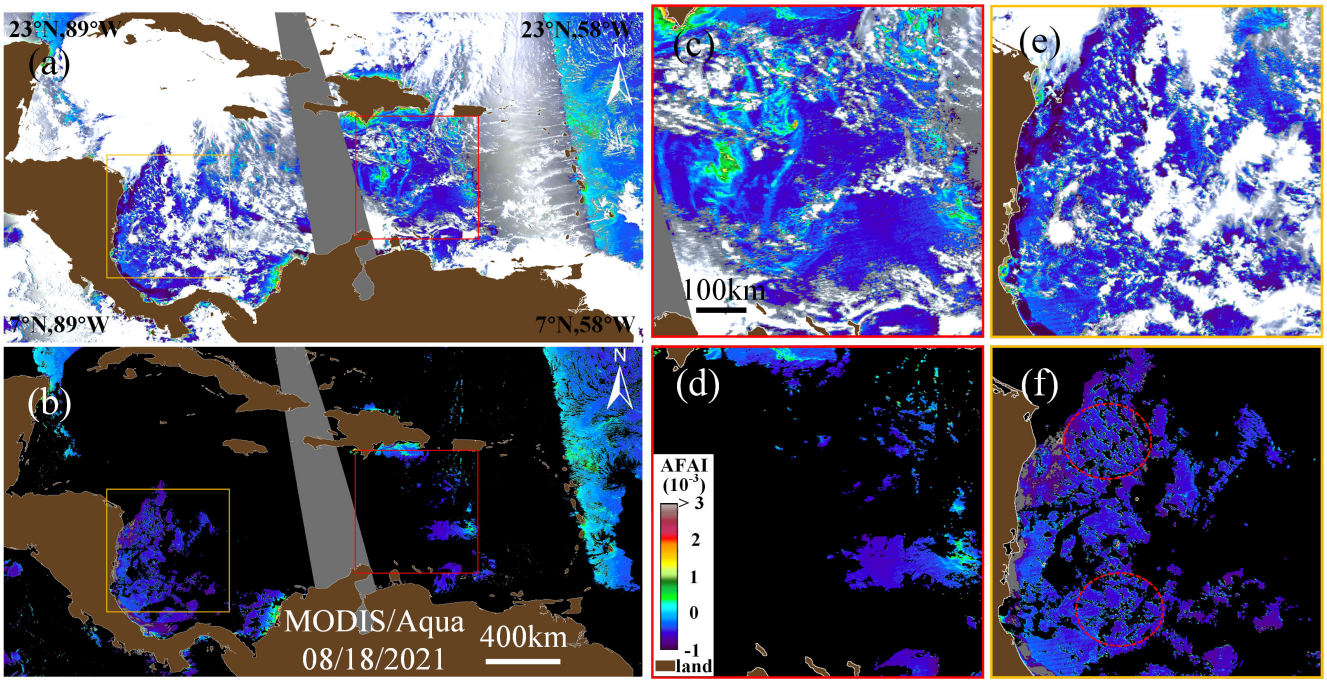


Fig. 1. SeaDAS CLDICE masking performance on MODIS. An example of the merged MODIS-Aqua AFAI image (a) on 18 August 2021 over Caribbean Sea and overlaid with cloud mask (b). In (a), (c), and (e), the invalid values on the AFAI images were replaced by RGB values to indicate the cloud locations. In (b), (d), and (f), black color indicates SeaDAS_cloud, and gray color indicates the invalid values or no data coverage on the AFAI image. (c) and (d) are the enlarged regions of the red boxes in (a) and (b), respectively. (e) and (f) are the enlarged regions of the yellow boxes in (a) and (b), respectively. SeaDAS_cloud overmasked valid pixels containing both water and *Sargassum* in (d) and failed to identify cloud shadow pixels in the red circle in (f). The brown color marks the land pixels.

cloud shadows are both found on the SeaDAS_cloud masked image. The imperfect masking not only induces uncertainties in biomass estimations, but also hinders bloom tracking and prediction. Considering that the bloom season over the Central Atlantic and Caribbean Sea usually occurs in the summer months when sufficient measurements are required, errors in glint masking may further induce large data gaps. Therefore, to make better use of the multispectral imagery to monitor the macroalgal blooms, and to reduce the biomass estimation uncertainties, improved cloud masking is needed.

Recently, advanced deep learning techniques have demonstrated satisfactory performance in cloud detection for terrestrial monitoring. For example, Segal-Rozenhaimer et al. [21] used a convolutional neural network (CNN) to detect clouds with the red, green, blue (RGB), and NIR band signals and showed robust performance on both WorldView-2 and Sentinel-2 images. In addition, Chen et al. [24] proposed a neural network classifier driven by extensive radiative transfer model simulations, which showed preliminary success for cloud detection on Geostationary Ocean Color Imager (GOCI) images [25]. Compared with the traditional methods that rely only on spectral signatures [11], [26], the deep learning network can use both spatial and spectral information to enhance cloud detection accuracy [22]. Therefore, it is promising that deep learning methods can be explored to establish robust cloud detection workflow on similar ocean remote sensing images.

To better monitor the floating macroalgal blooms dynamics, the objective of this article is to use deep learning techniques to accurately identify the cloud and cloud shadow pixels on MODIS imagery. To achieve that, the cloud detection algorithm was first described and the extraction accuracy was evaluated. Then, the number of valid measurements and the corresponding macroalgal amount derived from the deep-learning-based cloud masks were compared with these generated from the traditional methods. After that, their discrepancies in different seasons and locations were quantified and the possible reasons for their distinct performance were discussed. Finally, the applicability of the proposed approach was verified by applying it to similar VIIRS AFAI images.

II. Data and Methods

A. Satellite Data and Preprocessing

The Level-0 MODIS-Aqua data covering the Caribbean Sea (7°–23°N, 89°–58°W) from December 2020 to December 2021 were obtained from the U.S. National Aeronautics and Space Administration (NASA) Goddard Space Flight Center (http://oceancolor.gsfc.nasa.gov) and then processed to the Level-2 Rayleigh-corrected reflectance (Rrc) for each spectral band using the SeaDAS (version 8.2.0). The Rrc data were then mapped at 1-km spatial resolution in the equidistant rectangular projection. A total of 1742 5-min granules from MODIS-Aqua were downloaded and processed. The AFAI value of each pixel was then calculated using the related Rrc band following the equations below:

$$\mathrm{AFAI} = R_{\mathrm{rc,NIR}} - R'_{\mathrm{rc,NIR}}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:33:09 UTC from IEEE Xplore. Restrictions apply.

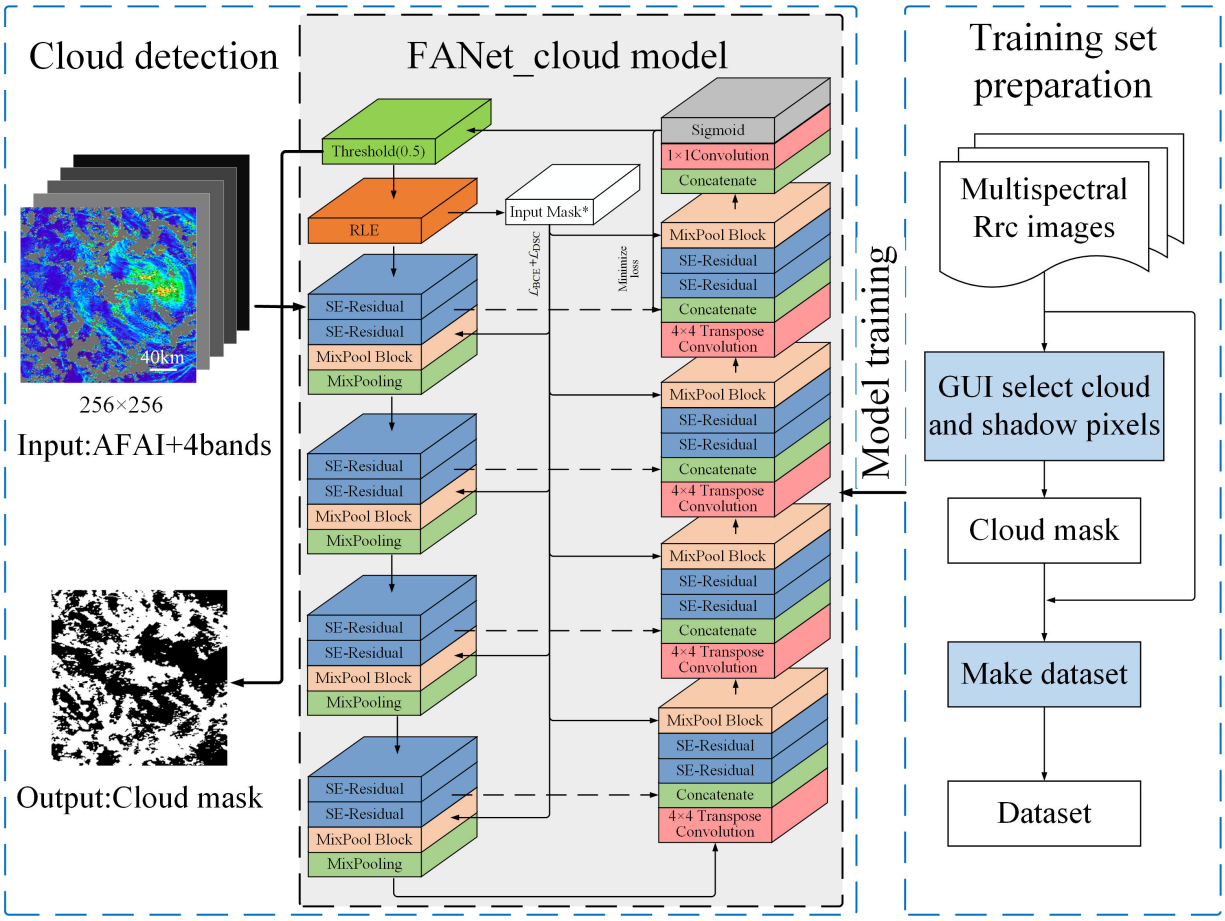


Fig. 2. Proposed cloud detection workflow on MODIS images.

$$R_{\mathrm{rc,NIR}} = R_{\mathrm{rc,RED}} + \left(R_{\mathrm{rc,LNIR}} - R_{\mathrm{rc,RED}}\right)\left(\lambda_{\mathrm{NIR}} - \lambda_{\mathrm{RED}}\right) / \left(\lambda_{\mathrm{LNIR}} - \lambda_{\mathrm{RED}}\right) \quad (1)$$

where $R_{\mathrm{rc},\lambda}$ represents the Rrc data of band λ, and λ_{RED}, λ_{NIR}, and λ_{LNIR} (LNIR, long NIR wavelength) represent 667, 748 and 869 nm, respectively.

A total of 2823 VIIRS 6-min granules' data [1386 from Joint Polar Satellite System-1 (VIIRS-JPSS1) and 1437 from Suomi National Polar-orbiting Partnership (VIIRS-SNPP)] in 2021 were also tested to show the applicability of the proposed cloud detection approach on similar sensors. VIIRS data were acquired from the same source as MODIS and also processed to Level-2 Rrc by SeaDAS. For VIIRS-JPSS1 AFAI, λ_{RED}, λ_{NIR}, and λ_{LNIR} represent 667, 746, and 868 nm, respectively. For VIIRS-SNPP AFAI, λ_{RED}, λ_{NIR}, and λ_{LNIR} represent 671, 745, and 862 nm, respectively.

B. Cloud Detection

A deep learning model named feedback attention network (FANet) was trained for cloud masking on MODIS multispectral images. Although the FANet model proposed by Tomar et al. [27] was first designed for biomedical image segmentation, it is generic enough and can apply to our study because cloud masking on satellite ocean color images is essentially an image segmentation task. Recent studies have successfully used similar deep learning methods initially designed for biomedical images for cloud masking on land remote sensing images [21], [22] and *Sargassum* extraction on ocean remote sensing images [16], [28] with satisfactory performance.

The architecture of the FANet is able to locate targeted features with a feedback attention mechanism that can unify the previous epoch mask with the feature map of the current epoch in training and provide attention to the learned feature maps at each convolutional layer. The complete network architecture, which constitutes a cycle between consecutive epochs, is shown in Fig. 2. The SE-Residual Block is designed to solve vanishing or exploding gradients problem, and MixPool Block is used to facilitate information feedback between consecutive epochs and pay hard attention to the learned features. FANet model has 7.72 million parameters and 94.75 GMac flops.

The model input of our FANet for cloud detection includes five bands, i.e., the AFAI, and four Rrc bands at blue (469 nm), green (555 nm), red (645 nm), and shortwave-infrared (SWIR, 1240 nm) wavelengths. The AFAI band is selected to discriminate the floating algae signals from the water pixels. The RGB and SWIR band reflectance values are effective for discriminating cloud edge and cloud shadow pixels and are typically included in cloud detection models [21], [26]. For comparison, experiment was also conducted

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:33:09 UTC from IEEE Xplore. Restrictions apply.

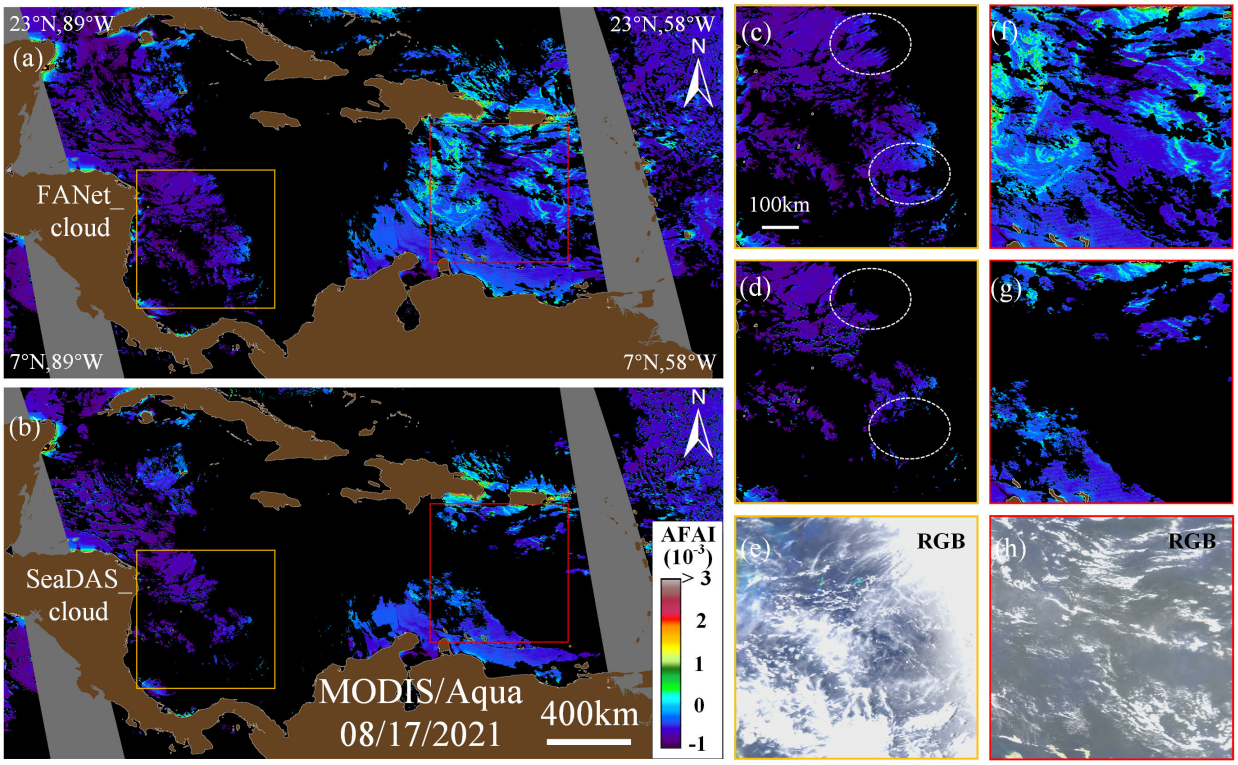


Fig. 3. Examples of less overmasking on the FANet_cloud compared with SeaDAS_cloud. Comparison between (a) FANet_cloud (this study) and (b) SeaDAS_cloud on one merged MODIS-Aqua image (August 17, 2021). (c) and (f) are the enlarged regions from (a). (d) and (g) are the enlarged regions from (b). (e) and (h) are the corresponding RGB images to show the performance of different cloud masks. SeaDAS_cloud overmasked water pixels in (d) and *Sargassum* pixels in (g). Black color indicates invalid values or masked data including clouds, cloud shadows, and strong sun-glint. Gray color indicates no observations. Overall, our FANet_cloud achieved less overmasking than SeaDAS_cloud [see the white circles in (d) and strong invalid mask in (g)].

using four bands including the RGB and SWIR bands and showed less satisfactory performance (see Fig. S1). The output of the FANet model is a cloud mask containing both cloud and shadow pixels. Our FANet model can learn the physical properties of cloud and cloud shadows from the RGB, AFAI, and SWIR band reflectance. The large training set can help it to differentiate clouds/cloud shadows from other confusing targets using their spatial and spectral signatures.

Our FANet model was trained for 150 epochs using Adam optimizer with an original learning rate of 0.0001. ReduceLROnPlateau callback was used to monitor the learning rate and would adjust it to 0.00001 for the optimal training performance. The loss function is a combination of binary cross-entropy and dice loss. The training images used in this study were divided into 256 × 256. Data augmentations, including flipping, rotation, and band stretching, were used before training. Another classic deep-learning-based image segmentation model (UNet) was also trained to compare the performance.

The main workflow of our cloud detection is illustrated in Fig. 2. The first step is to prepare samples for model training. A total of 27 merged images (https://github.com/mqleo/cloudmask.git) described in Section II-A were prepared for training. The four selected bands, AFAI, and the extracted cloud mask results were divided into 256 × 256 subimages to build the training set. Second, the prepared training set was used to train our FANet model, and the model parameters were adjusted to achieve better performance. Finally, the optimized FANet model was used to detect the cloud pixels on the input MODIS-Aqua images (see Section II-A).

As a data-driven deep learning method, the performance of the FANet model is dependent on the quality of manually-labeled training set. Here, the "true" cloud masks used in model training and validation were produced by the following procedures: 1) generating a preliminary cloud mask by combining the high SWIR band reflectance method used in [28] and the "whiteness" index method proposed by Gómez-Chova et al. [29]; and 2) ruling out false detections [false-positives (FPs) and false-negatives (FNs)] by referring to the AFAI and RGB images because cloud pixels are often diffuse features with high AFAI values and whitish color on RGB images. The manual corrections to rule out the false detections were conducted using a semiautomatic feature extraction graphic user interface developed by Wang and Hu [30].

The cloud detection accuracy was quantitatively evaluated by comparing to the SeaDAS_cloud and the N09CM-based cloud (N09CM_cloud) using the precision, recall, $F1$ score, and overall accuracy [31]. These metrics were defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \times 100\% \tag{2}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \times 100\% \tag{3}$$

$$F1\ \text{score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \times 100\% \tag{4}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:33:09 UTC from IEEE Xplore. Restrictions apply.

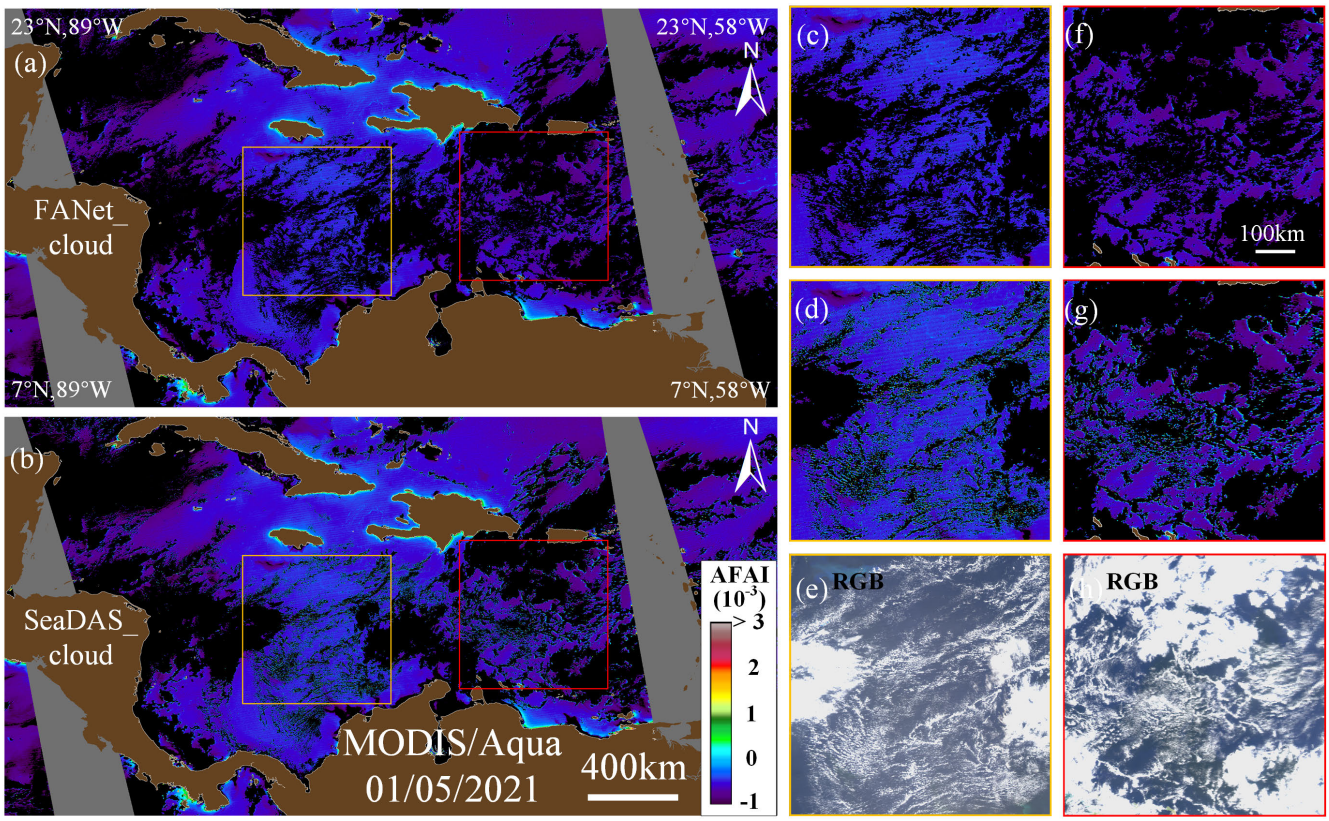


Fig. 4. Examples of less residual cloud shadows on the FANet_cloud compared with SeaDAS_cloud. Comparison between (a) FANet_cloud (this study) and (b) SeaDAS_cloud on merged MODIS-Aqua image (January 5, 2021). (c) and (f) are the enlarged regions from (a). (d) and (g) are the enlarged regions from (b). (e) and (h) are the corresponding RGB images to test the effect of cloud masks. SeaDAS_cloud undermasked broken cloud pixels in (d) and cloud shadow pixels in (g). Black color indicates invalid values or clouds/cloud shadows. Gray color indicates no data coverage. Overall, our FANet_cloud has less residual cloud shadows than SeaDAS_cloud.

$$\text{Overall Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \times 100\% \quad (5)$$

where true-positive (TP) indicates the number of correctly detected cloud and shadow pixels, true-negative (TN) denotes the number of correctly detected clear pixels (water or *Sargassum*-containing pixels unaffected by clouds), FP indicates the number of true clear pixels that are misidentified as cloud or cloud shadow pixels, and FN denotes the number of true cloud pixels that are misidentified as clear pixels.

C. Macroalgae Coverage and Biomass Estimations

The FANet model is generic enough to be readily used for floating *Sargassum* detection from satellite images. In our analyses, manual-labeled *Sargassum* samples were used to train another FANet model to detect *Sargassum*-containing pixels on MODIS images [32]. To quantify the impact of cloud masking on MODIS-derived floating algae abundances, the macroalgal coverages were estimated using the linear unmixing method described in [11] and then converted into biomass density using the AFAI-biomass model proposed by Wang et al. [12].

III. Results

A. Cloud Detection Performance: Qualitative and Quantitative

After targeted training, the FANet model was optimized to detect clouds on MODIS AFAI images. Our FANet-based cloud masking (FANet_cloud) was compared with SeaDAS_cloud on both the AFAI and RGB images to show its applicability for floating macroalgal detection.

Visually, the FANet_cloud tends to retain more valid observations and has fewer cloud and cloud shadow residuals that often degrade the subsequent macroalgae extraction and biomass estimations. Fig. 3 shows that FANet_cloud has less overmasking of valid measurements than SeaDAS_cloud. In the Caribbean Sea, SeaDAS_cloud, after being dilated outward for 1 pixel to reduce residual clouds, suffers from pronounced overmasking in summer when glint contaminations are strong. In this case, many valid observations will be discarded, increasing uncertainties in the macroalgal statistics derived using SeaDAS_cloud.

Fig. 4 shows that FANet_cloud has much less cloud and cloud shadow residuals than SeaDAS_cloud. These residuals on the SeaDAS_cloud can seriously contaminate the AFAI images [see Figs. 4(d) and (g)]. Visually, FANet_cloud is also spatially more consistent with real cloud distribution (see the corresponding RGB images). In the Caribbean Sea, the cloud/cloud shadow residuals frequently occur in winter when the bloom intensities were low. These false signals can increase uncertainties in the *Sargassum* coverage estimation during the early blooms.

To evaluate the cloud detection accuracy quantitatively, the FANet_cloud, SeaDAS_cloud, and N09CM_cloud (see performance in Fig. S2) were compared with the "true" cloud

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:33:09 UTC from IEEE Xplore. Restrictions apply.

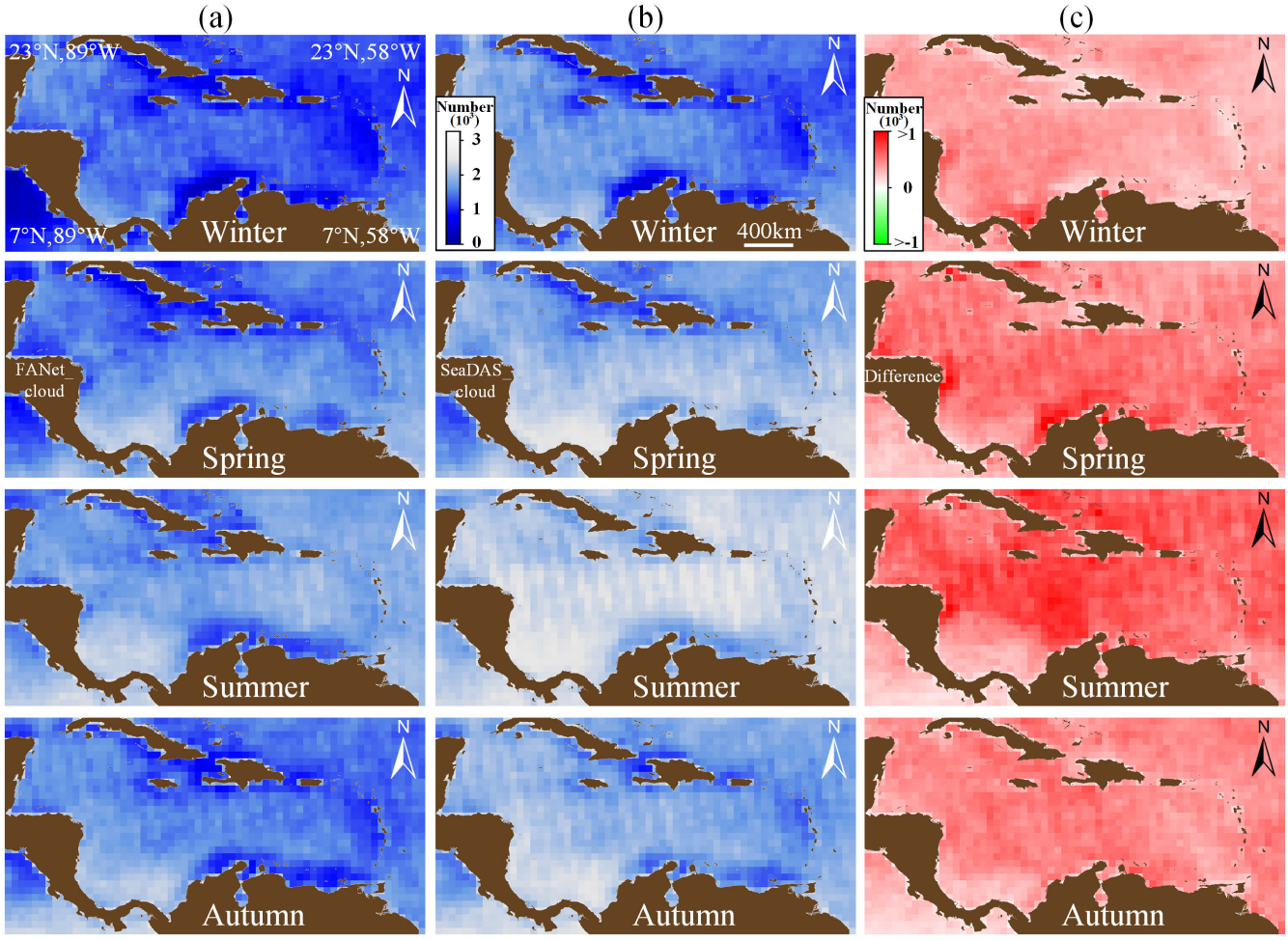


Fig. 5. Comparison of the seasonal mean invalid measurements using (a) FANet_cloud and (b) SeaDAS_cloud in 2021. The color indicates the mean number of invalid data in each 0.5° × 0.5° grid. SeaDAS_cloud products have more invalid measurements than FANet_cloud products. The images in (c) are the difference of the number of invalid measurements between FANet_cloud and SeaDAS_cloud (SeaDAS_cloud–FANet_cloud) in each season over the Caribbean Sea. Red color indicates the amount of SeaDAS_cloud overmasking than FANet_cloud in each 0.5° × 0.5° grid. Green color is the opposite. In (a) and (b), the whiter color indicates more invalid measurement. In (c), the reddish color indicates less valid measurements derived from SeaDAS_cloud.

masks (see Section II-B) to generate the accuracy metrics. 20 "true" cloud masks selected from all the months in 2021 were used as the validation data. The cloud detection accuracy on MODIS is listed in Table I.

The quantitative results indicate that FANet_cloud performed better than SeaDAS_cloud and N09CM_cloud. The overall accuracy and $F1$ score of FANet_cloud are both >95%. Most detection errors (either FPs or FNs) are caused by thin clouds that are hard to be discriminated. Nevertheless, the false detections of FANet_cloud are very low, and its FN rate is only 1.11%. The recall rates of FANet_cloud and SeaDAS_cloud are >90%, suggesting that most of the cloud pixels can be detected. Comparatively, the recall rate of FANet_cloud is high up to 98.89%, meaning that almost all the cloud and shadow pixels can be accurately masked. The precision of FANet_cloud is 91.72%, which is much higher than 80.26% of SeaDAS_cloud, suggesting that FANet_cloud has fewer FPs than SeaDAS_cloud. Although N09CM_cloud and SeaDAS_cloud showed similar performance, they both achieved lower precision and recall than FANet_cloud. A comparison of the performance of FANet and UNet is presented in Fig. S3. The higher cloud detection accuracies of FANet_cloud indicate that it has great potential for floating macroalgae detection from MODIS-like sensor data.

B. Impact of Cloud Masking on the Number of Valid Measurements

For the MODIS Aqua measurements collected in the Caribbean Sea in 2021, the seasonal mean invalid measurements derived using FANet_cloud and SeaDAS_cloud show significant differences. These invalid measurements include cloud, shadow, strong sun-glint, and invalid values in the AFAI-related reflectance bands. Although both the cloud masks have shown frequent cloud cover in the Caribbean Sea, which can be attributed to the existence of the Inter Tropical Convergence Zone (ITCZ) [33], the FANet_cloud [see Fig. 5(a)] shows more valid measurements than SeaDAS_cloud [see Fig. 5(b)]. Overall, FANet_cloud can retain 25.95% more valid measurements than SeaDAS_cloud. In summer, the total number of valid measurements retained by FANet_cloud is 44.41% more than SeaDAS_cloud. Since

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:33:09 UTC from IEEE Xplore. Restrictions apply.

TABLE I

ACCURACY ASSESSMENT OF THE CLOUD MASKING PERFORMANCE OF FANET_CLOUD, SEADAS_CLOUD, AND N09CM_CLOUD ON MODIS USING MANUAL SELECTED SAMPLES AS GROUND TRUTH

	Precision	Recall	F1 score	Overall accuracy
FANet_cloud	91.72%	98.89%	95.17%	**96.32%**
SeaDAS_cloud	80.26%	91.88%	85.68%	88.73%
N09CM_cloud	86.69%	87.91%	87.30%	90.61%

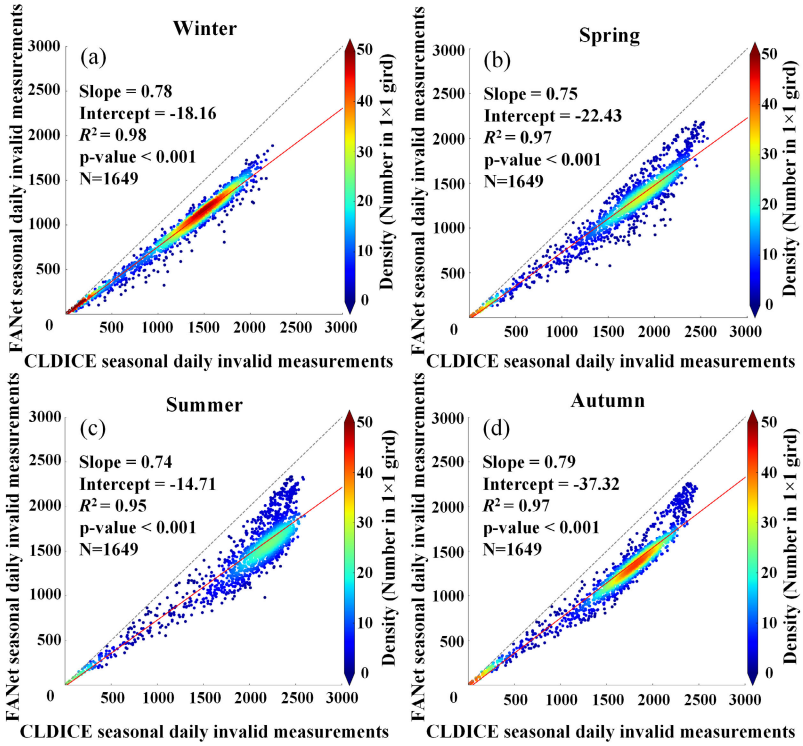


Fig. 6. Comparison of the daily mean invalid measurements in each season of 2021 estimated from FANet_cloud and SeaDAS_cloud. The dashed lines are the 1:1 reference lines, and the red solid lines are the linear regression lines. The color of the points indicates their density and the color bar is shown. There are 1649 points in each subgraph with the same number of 0.5° × 0.5° grids in the Caribbean Sea (without land).

SeaDAS_cloud does not effectively mask cloud shadows [as shown in Figs. 4(d) and (g)], it was dilated outward for one pixel in this comparison.

On their difference maps shown in Fig. 5(c), almost every grid has reddish color, suggesting that FANet_cloud consistently has more valid measurements than SeaDAS_cloud in the entire Caribbean Sea. Only a small number of white grids (representing similar cloud masking results) are primarily observed near the coasts. During spring and summer with heavy clouds, SeaDAS_cloud masks more clouds than FANet_cloud, suggesting that FANet_cloud can provide more valid measurements during the major *Sargassum* bloom seasons.

Despite the large discrepancies between FANet_cloud and SeaDAS_cloud, their invalid measurements have a strong positive correlation (see Fig. 6), with the coefficient of determination (R^2) exceeding 0.95 and p-values less than 0.001 in all the subfigures. All the points in the four subfigures fall below the dashed line, which means that FANet_cloud derived more valid measurements than SeaDAS_cloud in all the grids. The negative intercepts further emphasize the difference between the two cloud products. The slopes of the linear regressions are below 0.80. Similar to the above results, the slopes of Figs. 6(b) and (c) are about 0.75 and less than those in Figs. 6(a) and (d). This suggests that SeaDAS_cloud has more pronounced overmasking in spring and summer when more clouds exist.

Overall, these results indicate that the cloud covers derived by FANet_cloud and SeaDAS_cloud exhibit similar spatial patterns and temporal variations. Nevertheless, using the FANet model for cloud detection, more valid measurements can be effectively retained, especially during the summer bloom season. Therefore, the FANet_cloud is more suitable for alarming summer blooms and for generating long-term time series of floating macroalgal distributions from MODIS satellite images.

C. Impact of Cloud Masking on the Macroalgal Coverage Estimations

To assess the impact of cloud masking on macroalgal coverage estimations, *Sargassum* features extracted from the MODIS images in 2021 were masked with FANet_cloud and SeaDAS_cloud. The macroalgal coverage (% cover) in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:33:09 UTC from IEEE Xplore. Restrictions apply.

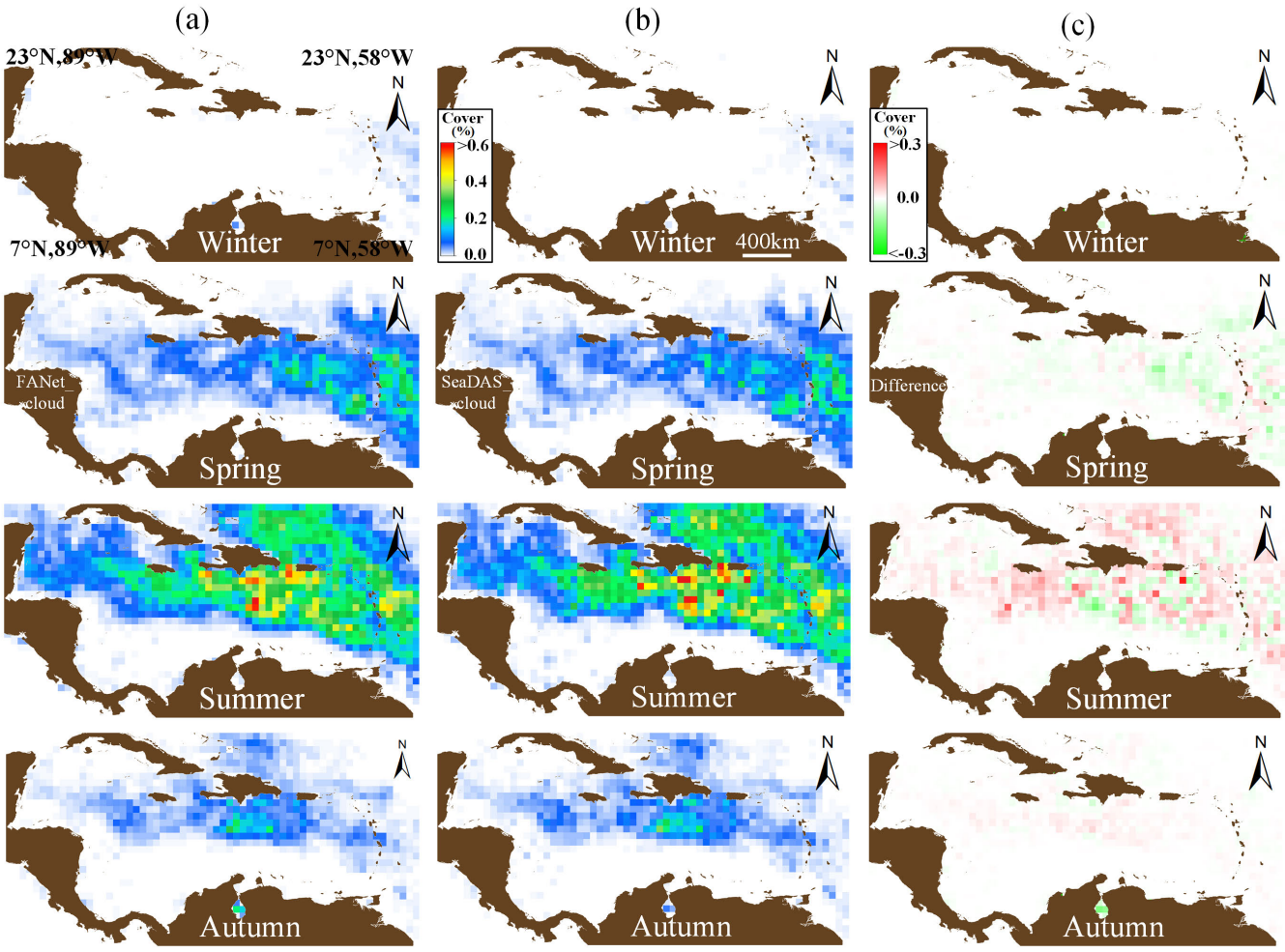


Fig. 7. Comparison of the seasonal mean *Sargassum* coverage in 2021 derived from MODIS-Aqua using (a) FANet_cloud and (b) SeaDAS_cloud. The color indicates % algae coverage in each 0.5° × 0.5° grid. The color of the grids in (c) are algae coverage differences between using FANet_cloud and SeaDAS_cloud derived in each season of 2021. Greenish color indicates more % algae coverage derived using FANet_cloud than using SeaDAS_cloud in each 0.5° × 0.5° grid. Reddish color is the opposite.

each 0.5° × 0.5° grid was determined following the workflow proposed in [11]. The available macroalgal coverage images collected during each season were used to generate the corresponding composite maps. Seasonal mean *Sargassum* distribution maps derived with FANet_cloud and SeaDAS_cloud show similar spatial patterns and temporal variations [see Figs. 7(a) and (b)]. Most *Sargassum* is distributed in the northern Caribbean Sea and large macroalgal coverages are found in the regions linked to the Central Atlantic. The *Sargassum* coverage increased from winter to summer and reached its peak in summer months.

To visualize the discrepancies of the seasonal mean *Sargassum* coverages derived with the two cloud masks, the differences between the two coverages were compared as shown in Fig. 7(c). In winter 2021, the *Sargassum* coverages obtained with both the cloud masks are similar. In spring 2021, higher *Sargassum* coverages were derived with FANet_cloud than SeaDAS_cloud. In summer and autumn 2021, *Sargassum* coverages derived with FANet_cloud are relatively lower than SeaDAS_cloud. These seasonal differences need to be carefully considered when using the two cloud masks.

Significant linear correlations were also found between the *Sargassum* coverages masked by FANet_cloud and SeaDAS_cloud products (see Fig. 8, $R^2 \geq 0.80$; p-value < 0.001). The slopes of subfigures are all close to 1. All the points are roughly distributed around the 1:1 line, indicating that *Sargassum* coverages derived with the two cloud masks are similar. Therefore, despite the large differences in the number of valid measurements (see Section III-B), their macroalgal coverage estimations are similar. This further confirms the previous conclusions that the mean-value composite (normalizing the number of valid measurements to estimate the mean macroalgal coverages) can effectively minimize the impacts of data gaps caused by clouds or cloud shadows and should provide reliable long-term macroalgal distribution time series [34].

Unexpectedly, the *Sargassum* coverages of the entire Caribbean Sea derived using FANet_cloud are ∼8% less than the values derived using SeaDAS_cloud for summer 2021. As discussed in Section III-B, FANet_cloud would retain more valid measurements, especially during summer. It is therefore natural to hypothesize that in summer the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:33:09 UTC from IEEE Xplore. Restrictions apply.

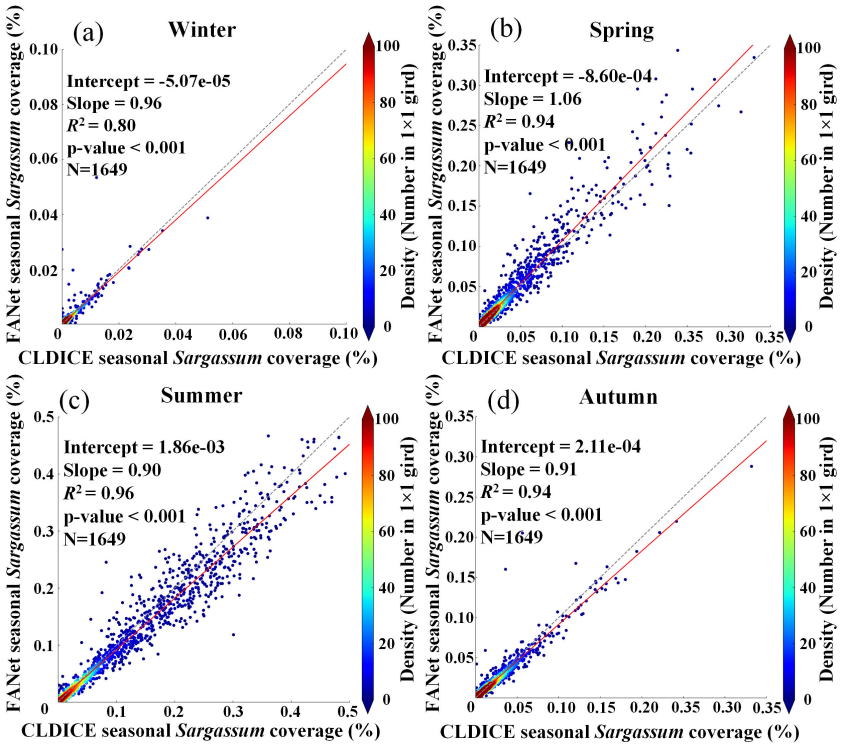


Fig. 8. Comparison of the seasonal mean *Sargassum* coverage in 2021 derived from MODIS-Aqua using FANet_cloud and SeaDAS_cloud. The dashed lines are the 1:1 reference line, and the red solid lines are the linear regression lines. The color of the points indicates their density, and the color bar is shown in subfigures. There are 1649 points in each subgraph with the same number of 0.5° × 0.5° grids in the Caribbean Sea (without land).

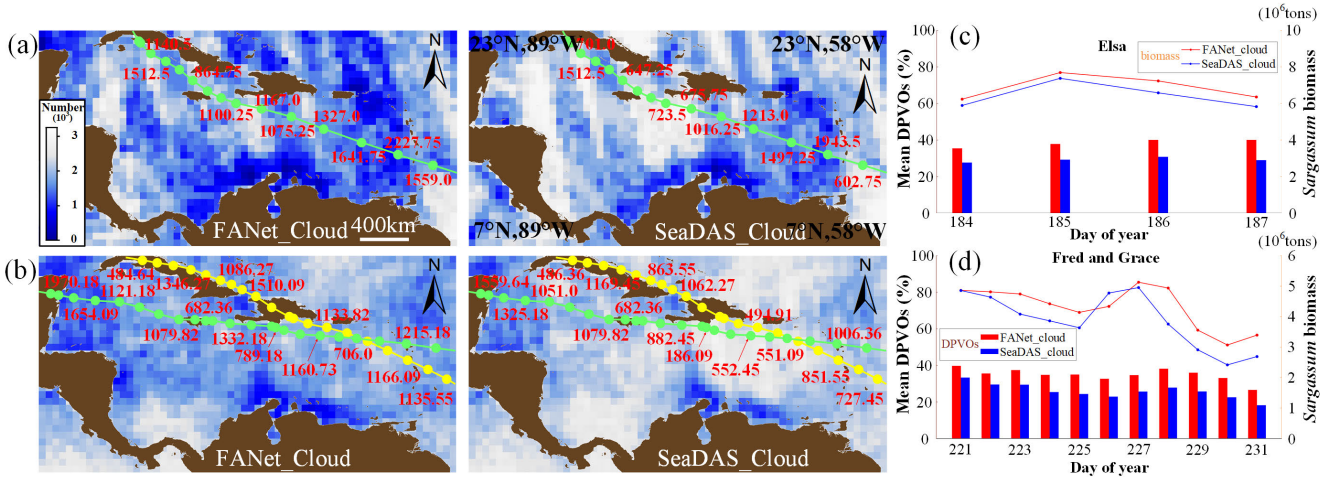


Fig. 9. Comparison of the FANet_cloud and SeaDAS_cloud derived cloud cover in hurricane periods of 2021. (a) and (b) show the cloud cover in Hurricane Elsa (2 July–6 July) and Tropical Storm Fred (yellow line) and Hurricane Grace (9 August–19 August). The color indicates the cloud cover in each 0.5° × 0.5° grid. The broken line in the picture is the hurricane track, and the points show the location of the hurricanes at 6-h intervals. The red labels are the number of valid measurements in the grid where the hurricane track points are located. Because there are no observed strips in MODIS during the day, the pattern in maps of cloud cover also exists strips. (c) and (d) are three-day mean *Sargassum* biomass and DPVOs calculated by overlaying FANet_cloud masking (red lines and bars) and SeaDAS_cloud (blue lines and bars) during two hurricane periods for the Caribbean Sea. The line charts show the change in *Sargassum* biomass. The bar charts show the mean DPVOs derived by two cloud masks. The horizontal coordinate is the day of year in each hurricane period.

larger valid coverages derived from FANet_cloud (where SeaDAS_cloud marked as invalid data) cover relatively more water than *Sargassum*, thus reducing the estimated mean composites.

To verify the above hypothesis, the percentages of *Sargassum* pixels under FANet_cloud and SeaDAS_cloud (without the common part) in summer were calculated. Fig. S4 shows the relationship between FANet_cloud and SeaDAS_cloud. The percentages of *Sargassum* pixels in B and C of Fig. S4(b) are 11.82% and 3.97% in all the summer images. This means that the valid coverages masked by FANet_cloud retain relatively more water than *Sargassum* compared with SeaDAS_cloud in summer, thus reducing the estimated mean composites.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:33:09 UTC from IEEE Xplore. Restrictions apply.

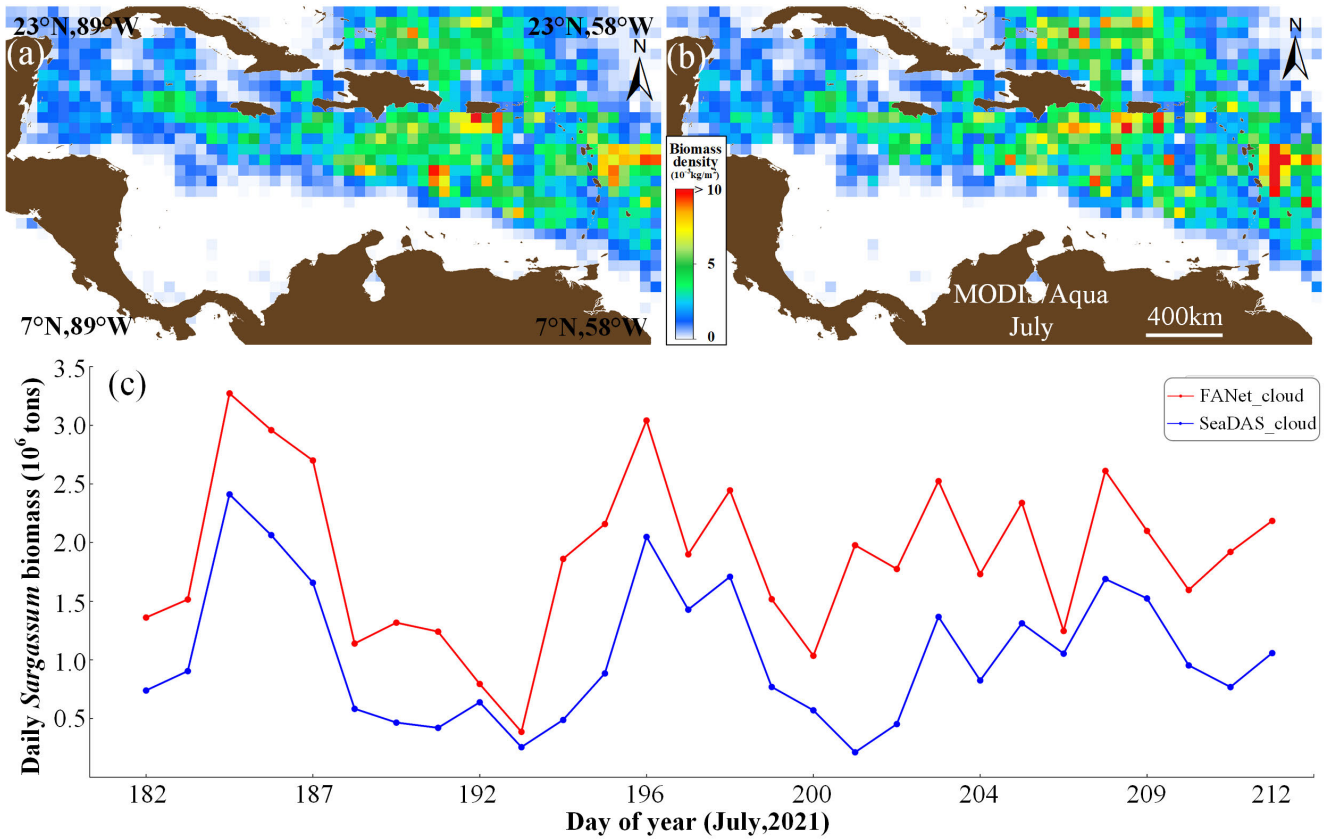


Fig. 10. Comparison of the daily *Sargassum* biomass observed in July 2021 in the Caribbean Sea. (a) and (b) show the monthly mean biomass density map in July derived from FANet_cloud and SeaDAS_cloud, respectively. (c) Observed *Sargassum* biomass was calculated by overlaying FANet_cloud (red lines) and SeaDAS_cloud (blue lines) on the daily MODIS-Aqua images in July 2021 over the Caribbean Sea. The horizontal coordinate is the day of year in July 2021.

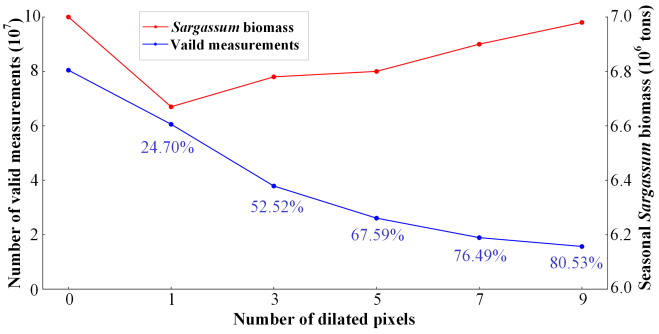


Fig. 11. Changes in the number of valid measurements (blue line) and estimated seasonal mean *Sargassum* biomass (red line) with the number of SeaDAS_cloud edge dilated pixels in summer of 2021. The horizontal coordinate is the number of dilated pixels for SeaDAS_cloud. The blue labels above the blue points represent the loss of valid measurements with an n-pixel dilation cloud compared with the original SeaDAS_cloud (no dilated).

In this study, two observed 2021 *Sargassum* biomass (see Fig. S5 for calculation) time series in the Caribbean Sea masked using FANet_cloud and SeaDAS_cloud were compared. The mean daily observed *Sargassum* biomass in 2021 over the Caribbean Sea derived with FANet_cloud is 0.64 million tons, while it is 0.40 million tons with SeaDAS_cloud (~60% more *Sargassum* biomass was derived with FANet_cloud). Although the differences in the seasonal mean *Sargassum* coverages derived from two cloud masks were not significant, there were still considerable differences in the daily observable *Sargassum* biomass.

IV. Discussions

A. Application for Retrieving Hurricane-Induced Sargassum Changes

Tropical cyclones (TCs), including hurricanes and tropical storms, frequently occur in the Caribbean Sea. The 2021 Atlantic hurricane season was the third-most active Atlantic hurricane season on record [35]. TCs often bring significant cloud cover in the affected area, making it hard to retrieve the hurricane-induced changes to the floating *Sargassum* from optical sensors. Previous efforts have reported that TCs can affect *Sargassum* abundance, but the difficulties can arise from the lack of measurements near the hurricane trajectories [36], [37]. In addition, since hurricanes frequently occur in summer when sun glint contamination is strong, glint masking can also affect the available measurements.

The comparison in Fig. 9 clearly shows that FANet_cloud retains more valid observations along the two hurricane tracks (across the entire Caribbean Sea) than SeaDAS_cloud. The daily percentage valid observations (DPVOs [38]) over the Caribbean Sea derived using FANet_cloud are higher than SeaDAS_cloud products in all the hurricane periods. Therefore, with more valid data, it is suggested that our FANet_cloud would better reveal the "true" *Sargassum*

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:33:09 UTC from IEEE Xplore. Restrictions apply.

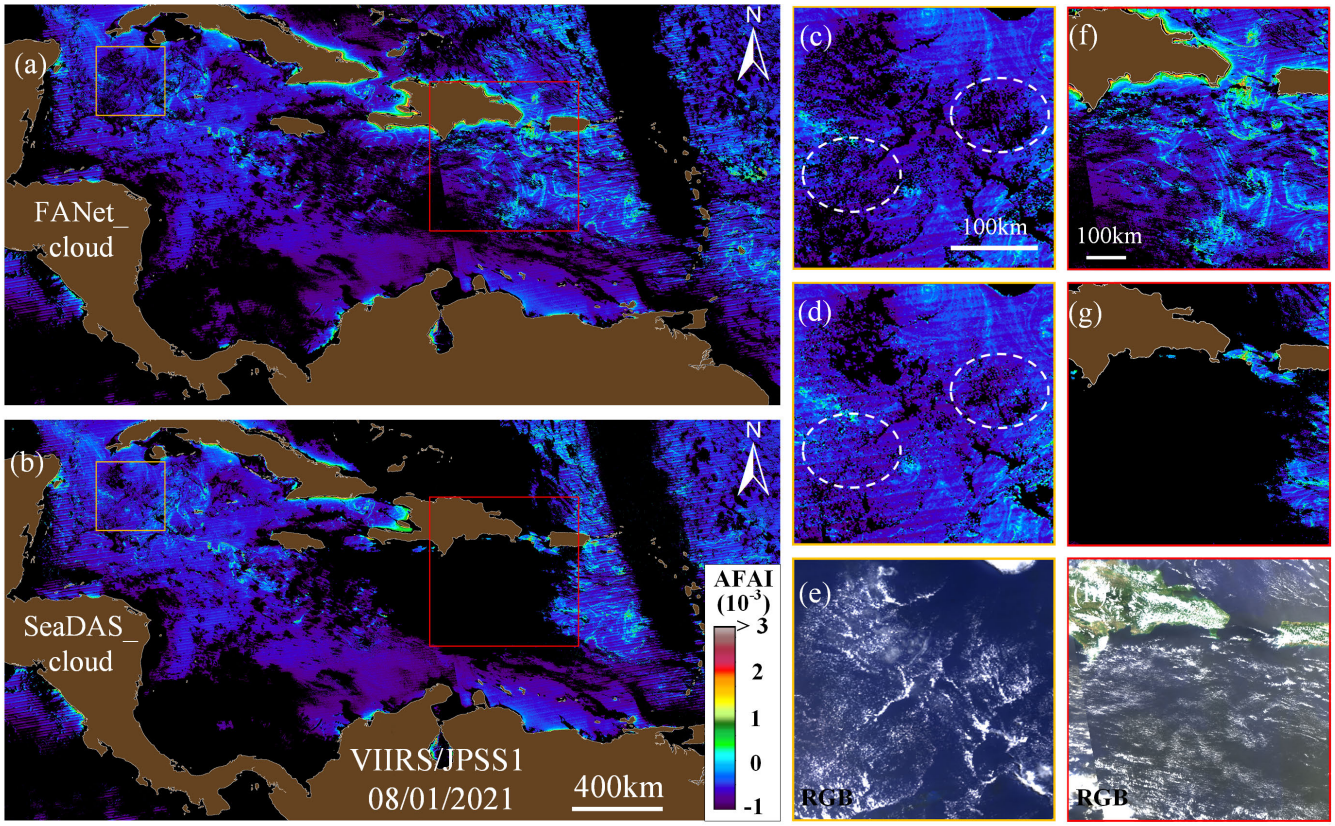


Fig. 12. Comparison between (a) FANet_cloud masking (this study) and (b) SeaDAS_cloud on merged VIIRS-JPSS1 image (August 1, 2021). (c) and (f) are the enlarged regions from (a). (d) and (g) are the enlarged regions from (b). (e) and (h) are the corresponding RGB images to test the effect of cloud masks. Black color indicates cloud masking including clouds, cloud shadows, and strong sun-glint. SeaDAS_cloud has a clear overmasking in (g), SeaDAS_cloud misses broken cloud and cloud shadow pixels in the white circles of (d). FANet_cloud performs well and is similar to real cloud in both the cases.

biomass changes before and after the hurricane overpass as shown in Figs. 9(c) and (d).

The mean *Sargassum* biomass values were compared for seven days before and after the transit of Elsa, Fred, and Grace. The seven-day mean biomasses after the TCs were lower than those before the TCs. The reduction even reached 55% after Fred and Grace passage that should be linked to the stronger impact with two TCs passing through the same area for longer time period. The reduced biomass is possibly caused by *Sargassum* sinking, natural decay of *Sargassum*, in addition to the increase in cloud cover and lack of observations [36], [39]. A suitable cloud masking will increase the probability to observe the real *Sargassum* biomass changes in response to extreme weather events.

B. Application for Detection of Sargassum Bloom Biomass Density During the Peak Month

With more valid measurements in the summer bloom season, it is also expected that the new FANet_cloud can help capture the maximum *Sargassum* biomass. According to the information from *Sargassum* Watch system (SaWS) [40], *Sargassum* biomass typically reached its peak around July in the Caribbean Sea. Figs. 10(a) and (b) show that the biomass densities of *Sargassum* obtained using two cloud masks have similar spatial patterns. Some of the *Sargassum* biomass density values are higher from SeaDAS_cloud especially in the dense regions, suggesting that cloud masking can bias macroalgal biomass density estimation during bloom months.

Despite the overall consistent mean distributions, the daily observed *Sargassum* biomass (see Fig. S5 for calculation) is much higher in the results masked by FANet_cloud as shown in Fig. 10(c). The observed *Sargassum* biomass estimations derived using FANet_cloud products are always higher than those using SeaDAS_cloud. This means that more *Sargassum*-containing pixels can be extracted using FANet_cloud, increasing the chance of observing the incoming *Sargassum* blooms to alert beaching events for coastal management.

On July 4, 2021, the total biomass obtained using two cloud masks in the peak month differs greatly. Our peak biomass is ~3.27 million tons in the Caribbean Sea, while the result from the SeaDAS_cloud is ~26% lower.

C. Impact of Cloud Masking Dilation

Cloud masking dilation can have a significant impact on floating macroalgal monitoring, even with small dilation. Since FANet_cloud can effectively mask out cloud shadows, outward dilations are not necessary, thus avoiding the loss of valid measurements, and the increase of *Sargassum* biomass estimation uncertainties caused by cloud-edge dilation.

In contrast, SeaDAS_cloud does not effectively mask cloud shadows and cloud edges. It is a common practice to dilate cloud-edge to ensure that those pixels do not impact

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:33:09 UTC from IEEE Xplore. Restrictions apply.

macroalgal extraction when using SeaDAS_cloud products [11], [14], [16]. However, the dilation strategy loses some of the valid observations including both "clean" water and *Sargassum*-containing pixels. To measure the loss of valid observations and the difference in macroalgal biomass estimations caused by cloud masking dilation, the valid number of measurements and seasonal mean *Sargassum* biomass in our study region with different dilation window sizes were quantified for summer 2021.

Fig. 11 shows the changes in the daily mean number of valid measurements and summer mean *Sargassum* biomass in the Caribbean Sea when the SeaDAS_cloud was dilated by 0, 1, 3, 5, 7, and 9 pixels outward. The number of valid measurements declines sharply as the dilation window size increases. When the number of dilated pixels is 3 (i.e., 7×7 window), the number of valid measurements is only half of that using the original SeaDAS_cloud. Even the most conservative 1-pixel dilation used in this study results in ∼25% less valid measurements. The seasonal mean *Sargassum* biomass tends to increase as the dilation window size increases. One exception is that dilating one pixel outward can lead to the decline in biomass, which is due to the fact that most of the cloud and cloud shadow residuals are effectively removed and thus reduced false detections in these locations. After that, *Sargassum* biomass increased as the valid observations decreased. This verified the hypothesis in Section III-C (the larger valid coverages cover relatively more water than *Sargassum*).

D. Application on Other Ocean Color Sensors

The FANet model performed well in cloud detection on MODIS images, both quantitatively and qualitatively, in the *Sargassum* time series analyses. Our proposed FANet_cloud can also be applicable to other similar multispectral moderate resolution ocean color sensor, such as VIIRS and OLCI. Using similar workflow, the cloud pixels on VIIRS images of entire 2021 were detected by the trained FANet model. Compared with SeaDAS_cloud products, FANet_cloud also achieved better performance on VIIRS (see Fig. 12) and can preserve a substantial number of valid observations, which include considerable *Sargassum*-containing pixels. FANet_cloud can also locate broken cloud and shadow pixels more precisely. More comparisons can be found in Figs. S6–S9 where the superior performance of FANet_cloud in reducing overmasking near sun-glint and cloud and cloud shadow residuals is illustrated.

V. Conclusion

In this study, a deep learning model was implemented for cloud masking on the multispectral ocean color images of MODIS. It provided more accurate cloud masking that can effectively reduce overmasking near sun-glint that frequently occur during the summer *Sargassum* bloom season and also have less cloud and cloud shadow residuals. The experimental results show that more valid measurements are retained and floating macroalgae detection and biomass estimation will be more reliable when using our FANet_cloud masking.

Comparisons of the valid retrieval during the hurricanes and bloom seasons suggested that using the FANet_cloud can better retrieve the *Sargassum* abundance changes during these critical periods. More importantly, the method is also applicable to other ocean color sensors. Following this work, multispectral ocean color sensors can retrieve more information on the surface oceanic features to maximize the measurement availability for monitoring floating macroalgae in the open oceans. In addition to *Sargassum* detection and monitoring, our generic FANet_cloud could also be optimized and used for *Ulva prolifera* monitoring near East China Sea from multispectral ocean color sensors.

Acknowledgment

This research was undertaken on the Supercomputing Center of Wuhan University (https://hpc.whu.edu.cn/). The authors thank National Aeronautics and Space Administration (NASA) for making the satellite data used in this article freely available.

References

[1] J. Gower, E. Young, and S. King, "Satellite images suggest a new *Sargassum* source region in 2011," *Remote Sens. Lett.*, vol. 4, no. 8, pp. 764–773, Aug. 2013, doi: 10.1080/2150704x.2013.796433.

[2] B. I. van Tussenbroek et al., "Severe impacts of Brown tides caused by *Sargassum* spp. On near-shore Caribbean seagrass communities," *Mar. Pollut. Bull.*, vol. 122, nos. 1–2, pp. 272–281, Sep. 2017, doi: 10.1016/j.marpolbul.2017.06.057.

[3] J. Gower and S. King, "Seaweed, seaweed everywhere," *Science*, vol. 365, no. 6448, p. 27, Jul. 2019, doi: 10.1126/science.aay0989.

[4] M. Wang, C. Hu, B. B. Barnes, G. Mitchum, B. Lapointe, and J. P. Montoya, "The great Atlantic *Sargassum* belt," *Science*, vol. 365, no. 6448, pp. 83–87, Jul. 2019. [Online]. Available: https://www.science.org

[5] R. E. Rodríguez-Martínez, E. Jordán-Dahlgren, and C. Hu, "Spatio-temporal variability of pelagic *Sargassum* landings on the northern Mexican Caribbean," *Remote Sens. Applications: Soc. Environ.*, vol. 27, Aug. 2022, Art. no. 100767, doi: 10.1016/j.rsase.2022.100767.

[6] C. Hu et al., "Sargassum coverage in the northeastern Gulf of Mexico during 2010 from Landsat and airborne observations: Implications for the deepwater horizon oil spill impact assessment," *Mar. Pollut. Bull.*, vol. 107, no. 1, pp. 15–21, Jun. 2016, doi: 10.1016/j.marpolbul.2016.04.045.

[7] Q. Xing et al., "Remote sensing of early-stage green tide in the Yellow Sea for floating-macroalgae collecting campaign," *Mar. Pollut. Bull.*, vol. 133, pp. 150–156, Aug. 2018, doi: 10.1016/j.marpolbul.2018.05.035.

[8] Q. Xing et al., "Monitoring seaweed aquaculture in the Yellow Sea with multiple sensors for managing the disaster of macroalgal blooms," *Remote Sens. Environ.*, vol. 231, Sep. 2019, Art. no. 111279, doi: 10.1016/j.rse.2019.111279.

[9] L. T. Bach, V. Tamsitt, J. Gower, C. L. Hurd, J. A. Raven, and P. W. Boyd, "Testing the climate intervention potential of ocean afforestation using the great Atlantic *Sargassum* belt," *Nature Commun.*, vol. 12, no. 1, p. 2556, May 2021, doi: 10.1038/s41467-021-22837-2.

[10] Q. Xing, L. Tosi, F. Braga, X. Gao, and M. Gao, "Interpreting the progressive eutrophication behind the world's largest macroalgal blooms with water quality and ocean color data," *Natural Hazards*, vol. 78, no. 1, pp. 7–21, Aug. 2015, doi: 10.1007/s11069-015-1694-x.

[11] M. Wang and C. Hu, "Mapping and quantifying *Sargassum* distribution and coverage in the central west Atlantic using MODIS observations," *Remote Sens. Environ.*, vol. 183, pp. 350–367, Sep. 2016, doi: 10.1016/j.rse.2016.04.019.

[12] M. Wang et al., "Remote sensing of *Sargassum* biomass, nutrients, and pigments," *Geophys. Res. Lett.*, vol. 45, no. 22, p. 359, Nov. 2018, doi: 10.1029/2018gl078858.

[13] J. Gower and S. King, "The distribution of pelagic *Sargassum* observed with OLCI," *Int. J. Remote Sens.*, vol. 41, no. 15, pp. 5669–5679, Aug. 2020, doi: 10.1080/01431161.2019.1658240.

[14] A. Ody et al., "From in situ to satellite observations of pelagic *Sargassum* distribution and aggregation in the tropical North Atlantic Ocean," *PLoS ONE*, vol. 14, no. 9, Sep. 2019, Art. no. e0222584, doi: 10.1371/journal.pone.0222584.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:33:09 UTC from IEEE Xplore. Restrictions apply.

[15] S. B. Hooker. (2003). *SeaWiFS Postlaunch Technical Report Series Volume 22, Algorithm Updates for the Fourth SeaWiFS Data Reprocessing*. [Online]. Available: http://www.sti.nasa.gov/STI-homepage.htm1

[16] C. Hu et al., "Mapping and quantifying pelagic *Sargassum* in the Atlantic Ocean using multi-band medium-resolution satellite data and deep learning," *Remote Sens. Environ.*, vol. 289, May 2023, Art. no. 113515, doi: 10.1016/j.rse.2023.113515.

[17] Q. Xing et al., "World's largest macroalgal blooms altered phytoplankton biomass in summer in the Yellow Sea: Satellite observations," *Remote Sens.*, vol. 7, no. 9, pp. 12297–12313, Sep. 2015, doi: 10.3390/rs70912297.

[18] W. Shi and M. Wang, "Green macroalgae blooms in the Yellow Sea during the spring and summer of 2008," *J. Geophys. Res., Oceans*, vol. 114, no. 12, Dec. 2009, doi: 10.1029/2009jc005513.

[19] K. Nordkvist, H. Loisel, and L. D. Gaurier, "Cloud masking of SeaWiFS images over coastal waters using spectral variability," *Opt. Exp.*, vol. 17, no. 15, p. 12246, 2009, doi: 10.1364/oe.17.012246.

[20] C. Hu, "A novel ocean color index to detect floating algae in the global oceans," *Remote Sens. Environ.*, vol. 113, no. 10, pp. 2118–2129, Oct. 2009, doi: 10.1016/j.rse.2009.05.012.

[21] M. Segal-Rozenhaimer, A. Li, K. Das, and V. Chirayath, "Cloud detection algorithm for multi-modal satellite imagery using convolutional neural-networks (CNN)," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111446, doi: 10.1016/j.rse.2019.111446.

[22] J. H. Jeppesen, R. H. Jacobsen, F. Inceoglu, and T. S. Toftegaard, "A cloud detection algorithm for satellite imagery based on deep learning," *Remote Sens. Environ.*, vol. 229, pp. 247–259, Aug. 2019, doi: 10.1016/j.rse.2019.03.039.

[23] Y. Li, W. Chen, Y. Zhang, C. Tao, R. Xiao, and Y. Tan, "Accurate cloud detection in high-resolution remote sensing imagery by weakly supervised deep learning," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112045, doi: 10.1016/j.rse.2020.112045.

[24] N. Chen et al., "New neural network cloud mask algorithm based on radiative transfer simulations," *Remote Sens. Environ.*, vol. 219, pp. 62–71, Dec. 2018, doi: 10.1016/j.rse.2018.09.029.

[25] Y. Fan et al., "OC-SMART: A machine learning based data analysis platform for satellite ocean color sensors," *Remote Sens. Environ.*, vol. 253, Feb. 2021, Art. no. 112236, doi: 10.1016/j.rse.2020.112236.

[26] M. Wang and W. Shi, "Cloud masking for ocean color data processing in the coastal regions," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 11, pp. 3196–3105, Nov. 2006, doi: 10.1109/tgrs.2006.876293.

[27] N. K. Tomar et al., "FANet: A feedback attention network for improved biomedical image segmentation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 9375–9388, Nov. 2023, doi: 10.1109/TNNLS.2022.3159394.

[28] M. Wang and C. Hu, "Satellite remote sensing of pelagic *Sargassum* macroalgae: The power of high resolution and deep learning," *Remote Sens. Environ.*, vol. 264, Oct. 2021, Art. no. 112631, doi: 10.1016/j.rse.2021.112631.

[29] L. Gomez-Chova, G. Camps-Valls, J. Calpe-Maravilla, L. Guanter, and J. Moreno, "Cloud-screening algorithm for ENVISAT/MERIS multispectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 12, pp. 4105–4118, Dec. 2007, doi: 10.1109/TGRS.2007.905312.

[30] M. Wang and C. Hu, "Extracting oil slick features from VIIRS nighttime imagery using a Gaussian filter and morphological constraints," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 10, pp. 2051–2055, Oct. 2015, doi: 10.1109/LGRS.2015.2444871.

[31] B. M. Sundheim, "Tipster/MUC-5: Information extraction system evaluation," in *Proc. 5th Conf. Message Understand.*, 1993, pp. 27–44.

[32] Y. Sun, M. Wang, M. Liu, and Z. Li, "Continuous *Sargassum* monitoring in the Caribbean Sea and Central Atlantic from multi-sensor satellite observations," *Remote Sens. Environ.*

[33] D. Wylie, D. L. Jackson, W. P. Menzel, and J. J. Bates, "Trends in global cloud cover in two decades of HIRS observations," *J. Climate*, vol. 18, no. 15, pp. 3021–3031, 2005.

[34] C. Hu et al., "Mapping Ulva Prolifera green tides from space: A revisit on algorithm design and data products," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 116, Feb. 2023, Art. no. 103173, doi: 10.1016/j.jag.2022.103173.

[35] NOAA. *National Oceanic and Atmospheric Administration (2021)*. Accessed: Mar. 6, 2023. [Online]. Available: https://www.noaa.gov/news-release/active-2021-atlantic-hurricane-season-officially-ends

[36] R. Sosa-Gutierrez, J. Jouanno, L. Berline, J. Descloitres, and C. Chevalier, "Impact of tropical cyclones on pelagic *Sargassum*," *Geophys. Res. Lett.*, vol. 49, no. 6, Mar. 2022, Art. no. e2021GL097484, doi: 10.1029/2021gl097484.

[37] Y. Li and D. Tang, "Tropical cyclone wind pump induced chlorophyll-a enhancement in the South China Sea: A comparison of the open sea and continental shelf," *Frontiers Mar. Sci.*, vol. 9, Dec. 2022, Art. no. 1039824, doi: 10.3389/fmars.2022.1039824.

[38] L. Feng and C. Hu, "Comparison of valid ocean observations between MODIS Terra and Aqua over the global oceans," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1575–1585, Mar. 2016, doi: 10.1109/TGRS.2015.2483500.

[39] N. F. Putman and C. Hu, "Sinking *Sargassum*," *Geophys. Res. Lett.*, vol. 49, no. 17, Sep. 2022, Art. no. e2022GL100189, doi: 10.1029/2022gl100189.

[40] C. Hu et al., "Sargassum watch warns of incoming seaweed," *Eos*, vol. 97, pp. 10–15, Dec. 2016.



Mingqing Liu received the B.S. degree from the Hefei University of Technology, Hefei, China, in 2020, and the M.S. degree from Wuhan University, Wuhan, China, in 2022, where he is currently pursuing the Ph.D. degree.

His main research interests include remote sensing of cloud detection and macroalgae monitoring.

Mengqiu Wang received the B.Sc. and M.Sc. degrees in remote sensing science and technology from Wuhan University, Wuhan, China, in 2012 and 2014, respectively, and the Ph.D. degree in marine science from the University of South Florida, Tampa, FL, USA, in 2018.

From 2018 to 2021, she was a Post-Doctoral Researcher with the Optical Oceanography Laboratory, College of Marine Science, University of South Florida. She is currently a Professor with the School of Remote Sensing and Information Engineering, Wuhan University. Her research interests include ocean color remote sensing and global change monitoring.

Dr. Wang serves as a Young Editorial Board Member for *Geo-Spatial Information Science*.

Yue Sun received the B.Sc. degree in surveying and mapping engineering from the Taiyuan University of Technology, Taiyuan, China, in 2016. She is currently pursuing the Ph.D. degree with Wuhan University, Wuhan, China.

Her research interests mainly include marine macroalgae monitoring, data gap filling, and spatiotemporal data fusion using remote sensing images.

Zhongbin B. Li received the B.Sc. degree in surveying engineering from Liaoning Technical University, Fuxin, China, in 2009, the M.Sc. degree from Tongji University, Shanghai, China, in 2012, and the Ph.D. degree in GIS and remote sensing from The Hong Kong Polytechnic University, Hong Kong, in 2016.

In 2015, he was a Visiting Ph.D. Student with the School of Geographical Sciences and Urban Planning, Arizona State University, Tempe, AZ, USA. Between 2016 and 2019, he was a Post-Doctoral Research Fellow with the Geospatial Sciences Center of Excellence, South Dakota State University, Brookings, SD, USA. Between 2019 and 2021, he was an Assistant Professor with the Center for Global Change and Earth Observations, Michigan State University, East Lansing, MI, USA. He is currently a Professor with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China. His research interests include global change, land change science, remote sensing, and sustainable development.

Dr. Li serves as a Young Editorial Board Member for *Geo-Spatial Information Science* and *Journal of Remote Sensing*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:33:09 UTC from IEEE Xplore. Restrictions apply.